

Survey of Metal-Insulator-Metal Diodes

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Abstract-This paper will be a survey of Metal-Insulator-Metal (MIM) diodes, as it relates to their RF-to-DC power conversion efficiency. As wireless communication systems expand into the terahertz frequency, traditional rectifiers become unusable due to nonlinearity concerns, junction charging and carrier transit times. MIM diodes provide an alternative method for charge transfer, using very low junction capacitance and quantum tunneling across an insulating barrier to harvest power and detect external voltage changes.

I. BACKGROUND & PRINCIPLES OF OPERATION

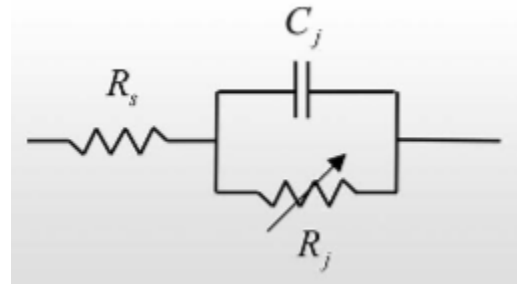
MIM diodes are made up of an extremely thin, nanometer-scale insulator between two conductive electrodes. The primary transport method of charge carriers through the insulator is through quantum tunneling. This mechanism allows for electrons to penetrate a sufficiently thin layer and create a current flow despite having a near zero bias voltage. The probability of an electron to tunnel through the insulator bandgap depends exponentially on the thickness of the insulator. This process produces a nonlinear current-voltage (I-V) characteristic. The secondary transport method is through thermionic emission (Schottky effect) in which charge carriers go over the insulator barrier. [2] This method usually only occurs when the

insulator is very large or the temperature is high, and for high frequency applications it is almost entirely quantum tunneling.

Traditional tunnel (Esaki) diodes also use the quantum tunneling effect to transport charge carriers, but instead they operate in the bandgap region, using a heavily doped p-n junction to cause inter-band electron tunneling [3]. The primary difference is that although both devices are nonlinear and can be represented through a Taylor series expansion of linear, rectification, and distortion terms, MIM diodes can rectify around a zero bias, passively converting RF signals to DC, while tunnel diodes use a biasing voltage.

II. RF-TO-DC CONVERSION EFFICIENCY

For analysis on energy harvesting, the equivalent MIM diode model can be seen in figure 1 [2].



This model includes a R_j , representing the zero bias resistance, in parallel with a C_j , for the zero

bias junction capacitance, which can go into the attofarad level allowing for terahertz operation, and with an R_s in series for the series resistance.

The rectification and theoretical harvester efficiency levels are defined by the Taylor series approximation of the circuit at a near-zero bias.

$$I(V) = \alpha_1 V + \alpha_2 V^2 + \alpha_3 V^3$$

Using equation 1, the terms can be divided into the linear ($\alpha_1 V$), rectification ($\alpha_2 V^2$), and distortion ($\alpha_3 V^3$) terms as the current response to a voltage input. For an RF input, equation 2:

$$V(t) = V_0 \cos(\omega t)$$

The DC component of the current that will be:

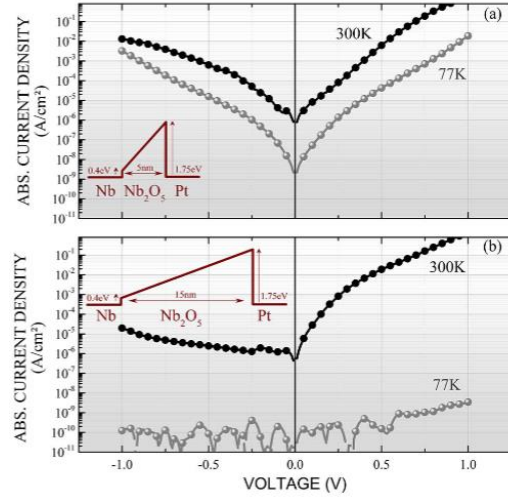
$$I_{DC} = \frac{\alpha_2}{2} V_0^2$$

This DC current represents the rectification without an applied bias voltage and instead is controlled by the coefficient of the curvature of the tunneling characteristic, α_2 . This is the theoretical harvester pickup, and to transform this into efficiency between an antenna and a load, the following equation 4 will be used:

$$\eta = \frac{P_{RL}}{P_{RA}} = \frac{I_{DC}^2 \cdot R_L}{\frac{V_0^2}{2 \cdot R_A}} = \frac{\alpha_2^2}{2} \cdot V_0^2 \cdot R_L \cdot R_A$$

This relationship between load resistance R_L , antenna resistance R_A , and the I_{DC} show the importance of proper matching networks in the system, as non-conjugate matching will greatly complicate the formulas.

Figure 2 below displays the current density to voltage characteristic of two 80 μ m MIM diodes, one with 5nm insulator thickness and the other with 15nm [2].



Although small, the current density values over voltage allow the MIM diode to be used as an energy-harvesting device.

III. SHORTCOMINGS

MIM diodes' I-V characteristic and tunneling probability depend on the shape of the tunnel barrier, specifically the height and thickness of the insulator. These characteristics also decide resistance and responsivity. A low resistance is essential for efficient coupling of the diode to the antenna and is done by keeping the barrier height low. But, a high responsivity is required for efficient small signal rectification [3]. These two fields are both affected in the same way by barrier height, specifically the height different between the left and right ends of the insulator. And so, as responsivity increases as will the resistance, which negates improvement in responsivity as the impedance becomes mismatched between the diode and antenna in a rectenna scheme. This creates a shortcoming in that, although the insulator is the first dependent variable an engineer can change to create the conditions desired, it is a constant tradeoff with each change.

Additionally, matching networks that aren't properly designed can cause major issues for rectenna performance, as seen in equation 4.

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